DEVENDU S DAS

ENGINEERING PROJECT PORTFOLIO

MECHATRONICS • EMBEDDED SYSTEMS • INDUSTRIAL IoT

Technical case studies presented with system block diagrams instead of project photographs.

PROFILE

Final-year B.Tech Mechatronics student at SRM Institute of Science and Technology, specializing in Industrial IoT and Systems Engineering. Interested in embedded systems, sensing, automation, connected devices, and system-level design.

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HOW TO READ THIS PORTFOLIO

The diagrams are conceptual summaries based on the project descriptions in the supplied resume, not verified final schematics. Replace or refine them when you have actual circuit diagrams, architecture models, code, or test evidence. No unverified performance results are claimed.

Engineering Profile

Technical focus

Embedded systems and firmware • IoT connectivity (BLE / Wi-Fi / MQTT) • real-time data acquisition • dashboard integration • control and automation basics • debugging and testing • system architecture and requirements.

Tools and skills listed in the resume

EMBEDDED & CONNECTIVITY	DATA & CONTROL
Embedded Systems & Firmware Development BLE / Wi-Fi / MQTT Git version control	Real-Time Data Acquisition Dashboard Integration Control Systems & Automation Basics Debugging & Testing Tools
DESIGN & ENGINEERING	**PROJECT TECHNOLOGIES**
SolidWorks / AutoCAD MATLAB / Simulink Problem Solving & Technical Collaboration	ESP32 / Arduino Node-RED / MQTT Python for plant-image analysis (project description)

EDUCATION

B.Tech, Mechatronics — Industrial IoT & Systems Engineering
SRM Institute of Science and Technology | 2023–2027

PROJECT SELECTION

The following case studies prioritize sensor integration, embedded control, IoT data flow, and systems design for a Mechatronics / Embedded Systems internship.

Automated Guided Vehicle (AGV)

Mechatronics Project | 2023–2024

PROJECT OVERVIEW

Developed and constructed a line-following AGV using IR sensors for path tracking, with microcontroller communication and real-time navigation control.

SYSTEM BLOCK DIAGRAM

Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
IR sensors, microcontroller units, and a motor/actuator system. Exact controller and motor-driver models are not specified in the resume.	Implemented master–slave communication, programmed navigation and control, and tested/debugged sensor-to-actuator response, as described in the resume.

SKILLS DEMONSTRATED

Sensor integration, embedded control logic, inter-controller communication, real-time response, and debugging.

Project status note: Use this page only for the elements you personally worked on; add actual controller details if you can verify them.

Smart Vertical Farming & Automated Fertigation

Mechatronics Project | 2026

PROJECT OVERVIEW

Designing an IoT-enabled vertical farming system for automated fertigation and environmental monitoring, connecting sensor inputs, dosing hardware, dashboards, and plant-growth observation.

SYSTEM BLOCK DIAGRAM

Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
Sensors listed: pH, TDS, turbidity, DHT22, CO■, BH1750, and ultrasonic. Other described elements include peristaltic pumps, solenoids, ESP32/Arduino, MQTT, Node-RED, Raspberry Pi camera, and Python image analysis.	The resume describes designing the monitoring, dosing, visualization, and image-analysis elements. Confirm which are implemented versus planned.

SKILLS DEMONSTRATED

Multi-sensor monitoring, embedded automation, fluid dosing, IoT visualization, and image-analysis concepts.

Project status note: The resume describes this project as in design. Keep planned functions clearly labelled until built and tested.

Smart Inventory Management

Mechatronics Project | 2025

PROJECT OVERVIEW

Developed an IoT inventory system with barcode-based equipment check-in/check-out and BLE notifications for high-value items crossing configured boundaries.

SYSTEM BLOCK DIAGRAM

Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
Barcode scanning, BLE notifications, and automated logging. Exact scanner, controller, database, and app platform are not specified.	Developed the inventory workflow and alert functionality, as described in the resume.

SKILLS DEMONSTRATED

Connected-device workflows, inventory event logging, BLE boundary-alert concepts, and system integration.

Project status note: Add the actual architecture or a redacted interface screenshot if available; do not show confidential inventory data.

Room Health Monitoring System

Mechatronics Project | 2025

PROJECT OVERVIEW

Developed an IoT-based environmental monitoring system that collects room-health parameters from multiple sensors and sends readings to a real-time dashboard.

SYSTEM BLOCK DIAGRAM

Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
Multiple environmental sensors and a real-time dashboard. Exact measured parameters, sensor models, network protocol, and dashboard platform are not specified in the resume.	Collected data from multiple sensors and sent updates to a dashboard for tracking room conditions, as described in the resume.

SKILLS DEMONSTRATED

Sensor data acquisition, IoT data flow, dashboard integration, and monitoring-oriented visualization.

Project status note: Name the exact parameters only if they match the actual system; add a real dashboard screenshot if you later obtain one.

Intelligent Detergent Dispenser

Systems Engineering Project | 2026

PROJECT OVERVIEW

Applied an MBSE approach to design an intelligent washing-machine detergent dispenser, from stakeholder/value analysis to subsystem architecture and requirements definition.

SYSTEM BLOCK DIAGRAM

Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
Stakeholder analysis; value-network and value-cost path analysis; sensing, fluidic, control, connectivity, and structural subsystems.	Defined functional, performance, safety, reliability, interface, environmental, and regulatory requirements; worked on architecture and verification/validation planning, per the resume.

SKILLS DEMONSTRATED

Requirements engineering, system decomposition, architecture definition, and verification/validation planning.

Project status note: This may be strongest as an architecture/requirements case study even without a physical prototype.

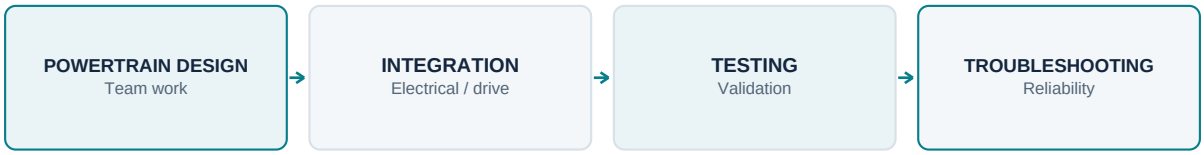
Electric Formula Racing Powertrain

4ZE Racing • Technical Member, Powertrain | 2025–2026

PROJECT OVERVIEW

Participated in a student team developing and testing an electric powertrain for a formula racing EV, including integration and troubleshooting of electrical and drivetrain systems.

SYSTEM BLOCK DIAGRAM



Conceptual flow for portfolio presentation; verify against the implementation before using as a final technical schematic.

TECHNOLOGIES / COMPONENTS	WORK DESCRIBED IN RESUME
Electric powertrain and drivetrain/electrical systems. Specific motor, inverter, battery, and control details are not provided in the resume.	The resume describes team participation in design, manufacturing, integration, testing, validation, troubleshooting, and optimization. State your own contribution precisely.

SKILLS DEMONSTRATED

Cross-functional engineering, integration, test-focused problem solving, and reliability-oriented troubleshooting.

Project status note: Use a conceptual system diagram only as an overview; do not present it as the exact vehicle wiring or powertrain schematic.

Engineering Experience

KELTRON — Kerala State Electronics Development Corporation Limited

Student Intern | December 2024 | Dharmasala, India

• Gained hands-on exposure to electronics manufacturing processes.
• Supported quality testing and assembly of capacitors and resistors.
• Contributed to production-data documentation and troubleshooting of test failures.
• Worked with team members on basic automation and measurement work.

Before sharing with recruiters

- Confirm the status of each project: designed, built, tested, or planned.
- Add exact hardware models and software tools only when you can verify them.
- Include real measurements, test evidence, code links, or actual diagrams only when available.
- Clarify your individual contribution on team projects.
- Review all contact details and remove any confidential or restricted material.

CONTACT

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Thank you for reviewing my portfolio.